

Quantitative Back Testing

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Abstract

We compare portfolio performance using cumulative returns, maximum drawdown, Sharpe ratio, Sortino ratio, and ARIMA-based time-series forecasting. Early analysis focuses on whether defensive or diversified portfolios remain more stable during periods of market stress compared to growth-heavy and benchmark portfolios



Introduction & Motivation

Problem & question: This project evaluates how different asset allocations behave across normal market conditions, growth periods, and financial crises. How can portfolio optimization help stakeholders construct portfolios that preserve return potential while reducing drawdowns during periods of financial crisis and market uncertainty?

Impact & stakeholders: This project is designed for stakeholders who need to evaluate portfolio risk beyond simple return.



Background & Related Work

Modern portfolio theory argues that diversification can reduce risk by combining assets with different return patterns. However, crisis periods often challenge diversification because correlations between assets may rise during market stress. Risk-adjusted measures such as the Sharpe ratio and Sortino ratio provide a more complete view of performance than return alone because they account for volatility and downside risk.

This project builds on portfolio theory, backtesting, crisis-period analysis, and time-series modeling. ARIMA is included as a statistical forecasting method to test whether portfolio return behavior becomes less predictable during stressful market periods.

References to include in final poster:

- Markowitz, H. Portfolio Selection.
- Sharpe, W. Capital Asset Prices and Risk-Adjusted Return.
- Sortino, F. Downside Risk Framework.
- Box and Jenkins / ARIMA time-series modeling.
- Taleb, N. Antifragile: Things That Gain From Disorder
- Yahoo Finance / [yfinance](#) documentation.

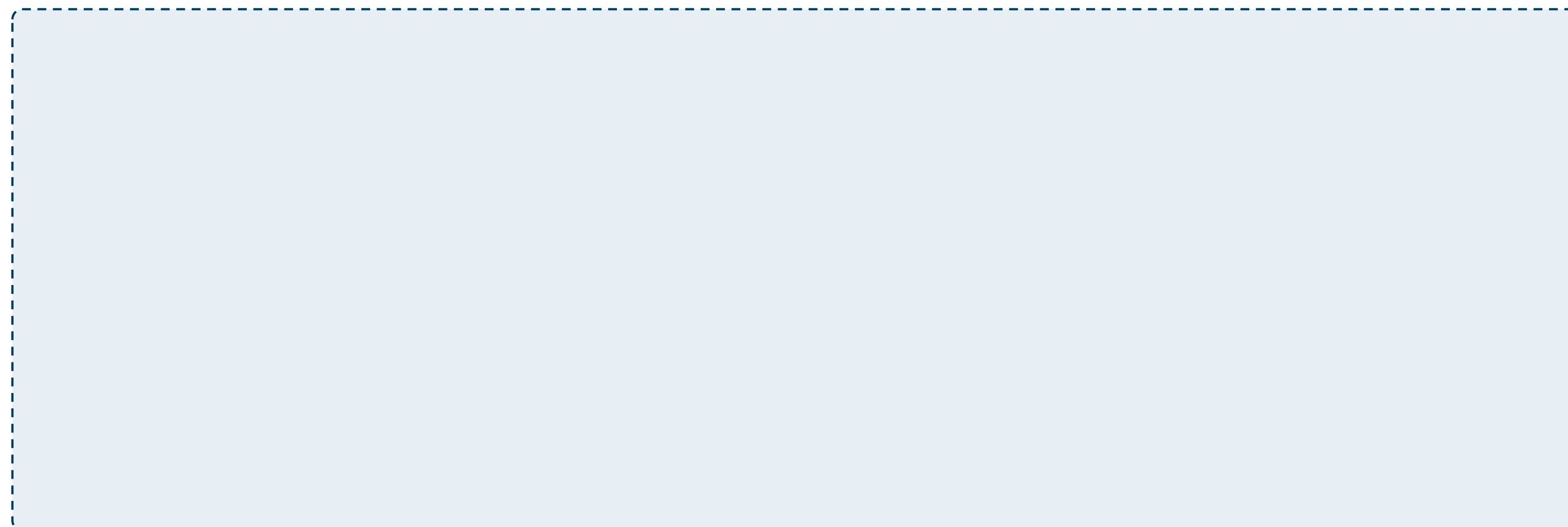


Data, Engineering & Ethics

Sources & scale: The project uses historical daily market data from Yahoo Finance accessed through the [yfinance](#) Python package. The M2 data scope is frozen through **2026-06-28**. No additional market data will be collected after this milestone.

Pipeline: Ticker list + portfolio weights -> [yfinance](#) price pull -> adjusted prices -> daily returns -> weighted portfolio returns -> crisis labels -> performance metrics -> visualizations

Ethics & responsible use: All data used in this project are public financial market data. The project does not use personally identifiable information, private investor records, or restricted financial data.



Methods & Evaluation

Approach: This project combines backtesting, risk analysis, and statistical time-series modeling.

Validation: Each portfolio is evaluated over the same historical period using daily returns. Portfolio returns are calculated by applying portfolio weights to individual asset returns.

Risk and Performance Metrics

The project evaluates each portfolio using:

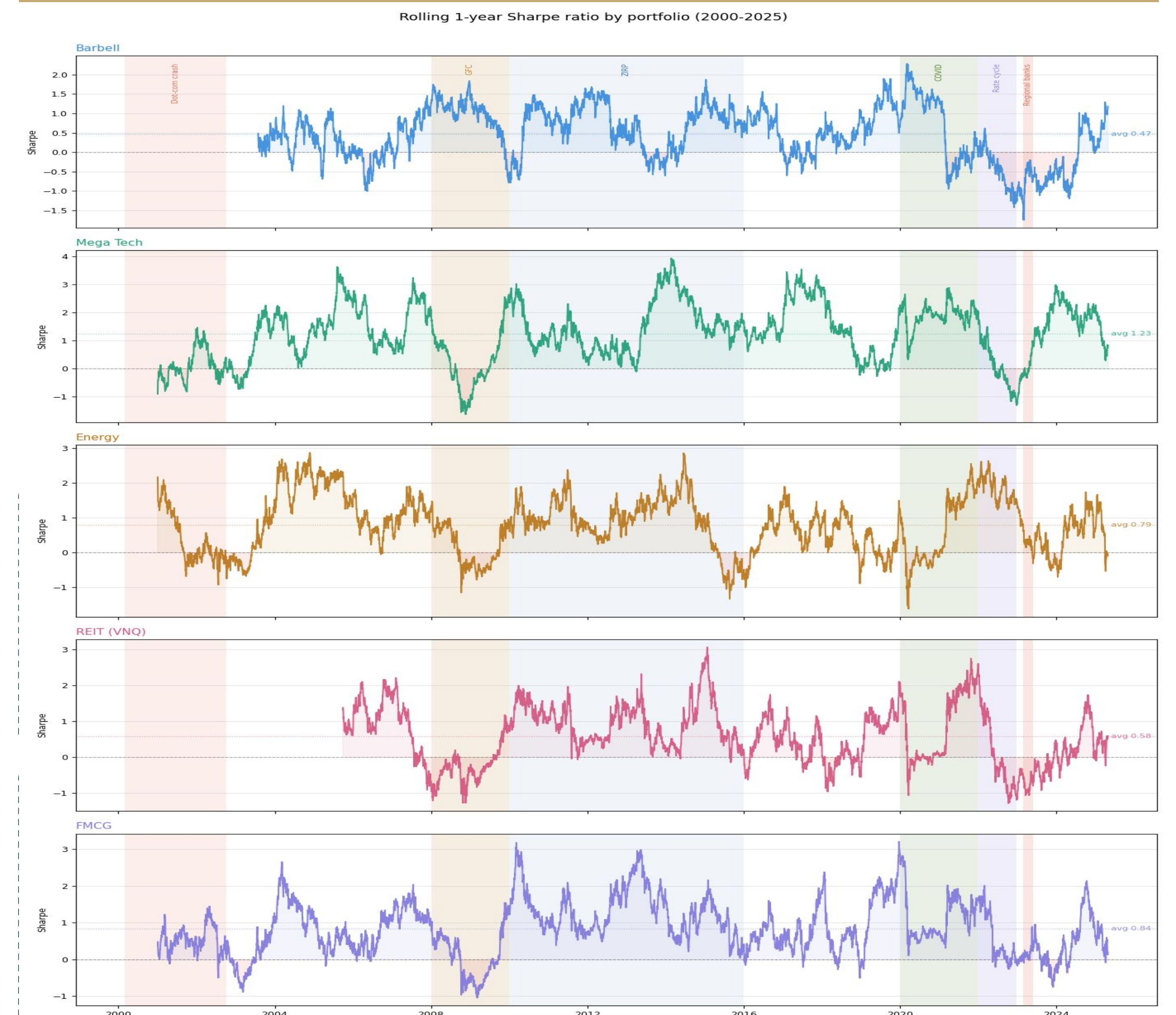
- Cumulative return
- Annualized volatility
- Sharpe ratio
- Sortino ratio
- Crisis-period return
- Recovery behavior
- Benchmark comparison

ARIMA Statistical Method

ARIMA is used as a statistical time-series forecasting method on portfolio-level return series. The purpose is not to predict the stock market perfectly, but to test whether some portfolios are more stable or predictable than others.



Results & Visualizations



Discussion, Limitations & Conclusions

Discussion: The project evaluates portfolio resilience from multiple angles rather than relying on return alone. A portfolio may generate high cumulative returns but still expose stakeholders to large drawdowns during crises. Similarly, a defensive portfolio may reduce downside risk but underperform during growth periods.

The combination of backtesting, drawdown analysis, Sharpe and Sortino ratios, benchmark comparison, and ARIMA forecast error gives a broader view of portfolio behavior under uncertainty.

Limitations & future work: Historical performance does not guarantee future performance.

- Backtesting can create false confidence if results are overinterpreted.
- Selected assets may introduce survivorship bias.
- Some assets have shorter histories due to IPO dates or fund creation dates.
- The VIX is an index, not a directly investable asset.
- Yahoo Finance data may revise adjusted prices or change ticker availability.
- ARIMA is a simple statistical forecasting baseline and may not capture nonlinear crisis behavior.

Conclusions: Portfolio resilience should be evaluated using multiple dimensions of performance, including return, volatility, drawdown, risk-adjusted return, and forecast stability.



Reproducibility & References

Repository: [<https://github.com/your-org/your-project>] · Key dependencies: [Python 3.x, pandas, scikit-learn, ...]

References: [1] Author, Title, Venue, Year. [2] ... (keep citations legible; QR to the repo is optional)

[QR:
optional]